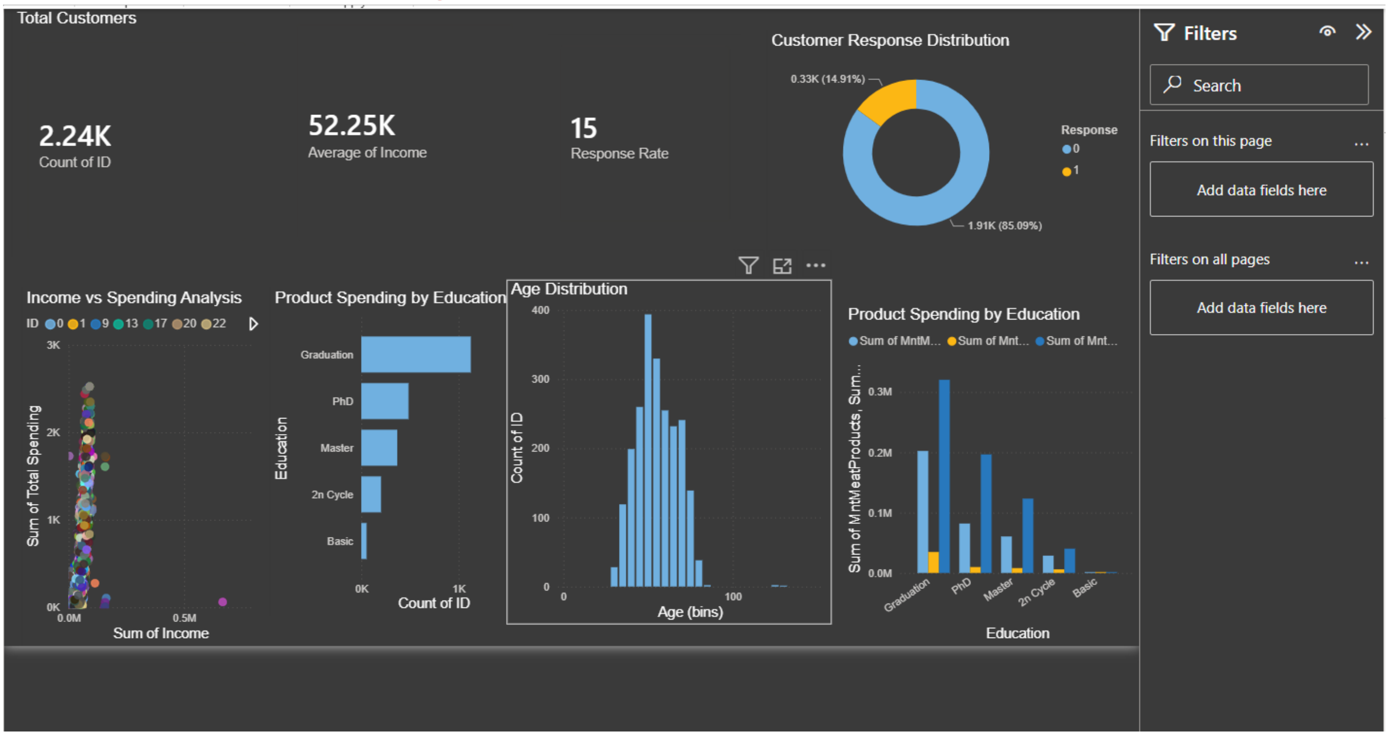
Marketing Campaign Data Storytelling & Statistical Validation

Task 4 Professional Report

Intern: Surya Prakash Sharma
Organization: ApexPlanet Software Pvt. Ltd.

This report presents an end-to-end analysis of a marketing campaign dataset using Python, Power BI and statistical hypothesis testing. The objective is to convert raw customer data into actionable business insights and validate those findings using statistical methods.

Interactive Power BI Dashboard



Dashboard Summary

The dashboard provides a consolidated business view including total customers, average income, campaign response rate, age distribution, education-wise product spending and income versus spending analysis. These visuals help decision makers quickly identify customer behaviour and marketing opportunities.

Business Insights

Key Findings
• Total Customers: 2,240
• Average Income: ■52,247
• Average Spending: ■605.8
• Campaign Response Rate: 14.91%

Interpretation
The dashboard indicates that customers with higher income generally spend more across product categories. Graduates contribute the highest purchasing volume. Campaign response remains comparatively low, suggesting that future campaigns should focus on customer segmentation and personalized targeting. Age distribution reveals that the majority of customers belong to the middle-age segment, representing the most valuable customer base.

Hypothesis Testing

Null Hypothesis (H■): Income has no significant effect on customer spending.
Alternative Hypothesis (H■): Income has a significant effect on customer spending.

Independent Two Sample T-Test Result:
T Statistic = 52.964
P Value = 0.000

Since the p-value is less than 0.05, the null hypothesis is rejected. Therefore, income significantly influences customer spending.

Business Recommendations

1. Target premium offers toward high-income customers.
2. Develop personalized campaigns for different customer segments.
3. Improve campaign response through customer profiling.
4. Monitor KPIs using Power BI dashboards.
5. Continuously validate business assumptions with statistical testing.

Future Scope
Future work may include predictive machine learning models, customer lifetime value prediction, recommendation systems and automated marketing analytics dashboards.

Conclusion

This project demonstrates the complete workflow of modern data analytics by integrating Python, Power BI, and statistical validation. The dashboard effectively communicates business performance, while hypothesis testing provides confidence in the findings. These insights enable organizations to improve campaign effectiveness, customer targeting, and marketing ROI.

Tools Used

Python • Pandas • NumPy • Matplotlib • Seaborn • SciPy • Power BI • Microsoft PowerPoint • GitHub